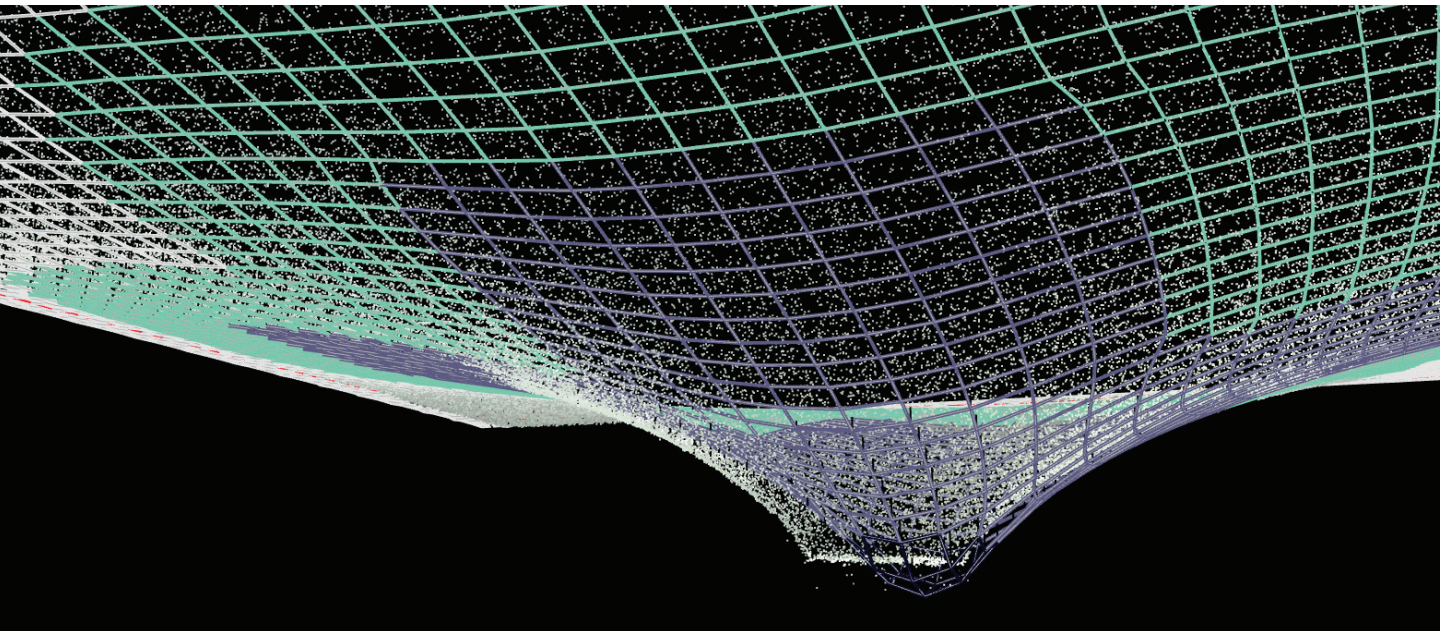
Simulation and Calibration of Graded Knitted Membranes

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ABSTRACT

The grading of knit changes its geometrical performance and steers membrane expansion. However, knit possesses challenges of material predictability and digital simulation, due to its multiscalar complexity and anisotropic properties. Taking as a challenge the lack of digital solutions incorporating CNC-knit performance into the design model, this paper presents a novel approach for the design-integrated simulation of graded knit, informed by an empirical dataset analysis in combination with genetic optimization algorithms.

Here the simulation design tool reflects the differences of industrially knitted textile panel behavior through digital mesh grading. Diversified fabric stiffness is achieved by inter-twining the yarn into variegated stitch types that steer the textile expansion under load. These are represented digitally as zoned quad meshes with each segment assigned a stiffness value.

Mesh stiffness values are optimized by minimizing the distance between the point clouds and the digital mesh, which are documented through deviation colored maps.

This work concludes that design properties—pattern topology, stitch ratio, pattern density—play an important role in textile panel performance under load. Stiffness values derived from the optimization are higher for shallower designs and lower for the deeper cones.

1 Fragment from the simulation digital environment, where the point cloud data is overlapped with the mesh-based geometry of a knitted membrane.

INTRODUCTION

In recent years numerous projects (Gupta et al. 2020; Popescu et al. 2020; Ramsgaard Thomsen et al. 2015, 2019) have displayed the potential of CNC knits as a next-generation material for textile architecture in its integrated detailing, material differentiation, and sustainable production. Knit grading changes the geometrical performance of the textile, allowing it to steer the membrane expansion in the predefined areas under load. However, knits possess challenges of material understanding, predictability, and digital simulation, due to their multi-scalar complexity and anisotropic properties.

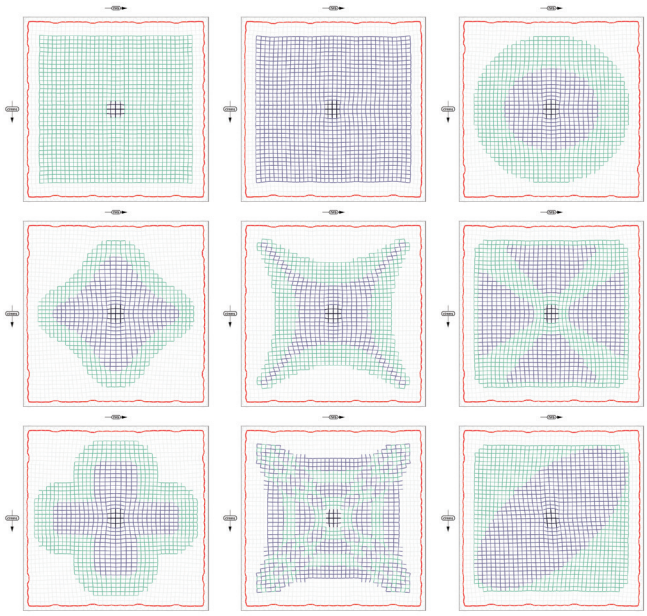
Currently, there are no digital solutions that would integrate the simulation of CNC-knit performance into a design model to specify fabrication data. Instead, the workflow is solved through a laborious sample-based prototyping, which inhibits scaling up and accessibility of knit application in architecture.

This paper presents a novel approach for the design-integrated simulation of CNC knits, informed by an empirical dataset analysis with the help of genetic optimization algorithms.

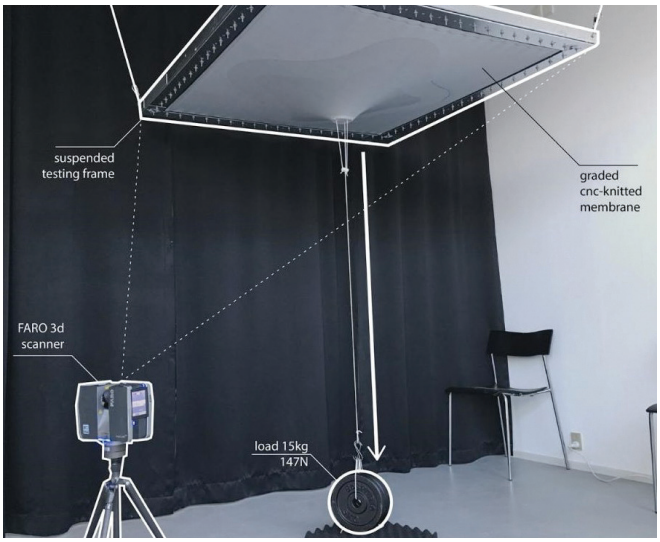
Existing simulation methods from fashion and computer graphics focus on human body templates, without taking into account the material stretch; rather, they represent fabric being draped over the mannequin (Kaldor, James, and Marschner 2008; ShimaSeiki 2007; STOLL 2007; Yuksel et al. 2012). Often these models are not connected to the fabrication machinery logic, although the behavior of the fabric is very realistic.

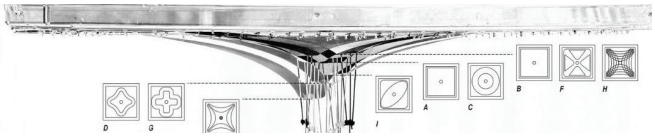
The application of knit in architecture aims to use fabrics under tension, where the geometrical performance and precision are highly crucial (Tamke et al. 2016). However, precise textile engineering methods of modeling through colliding stitches become computationally infeasible when scaling up (Kyosev and Renkens 2010). Architectural domain tools for woven membrane form-finding, although scalable, miss graded properties integration across the surface. Thus, the digital models for knitted membranes should reflect the differentiated surface properties of knit as these affect the geometry and material performance. Therefore, it is necessary to find ways to integrate graded properties of knit into design models (Ramsgaard Thomsen et al. 2016) in order to gain material understanding, inform its composition for fabrication, and promote the use of knit in architecture.

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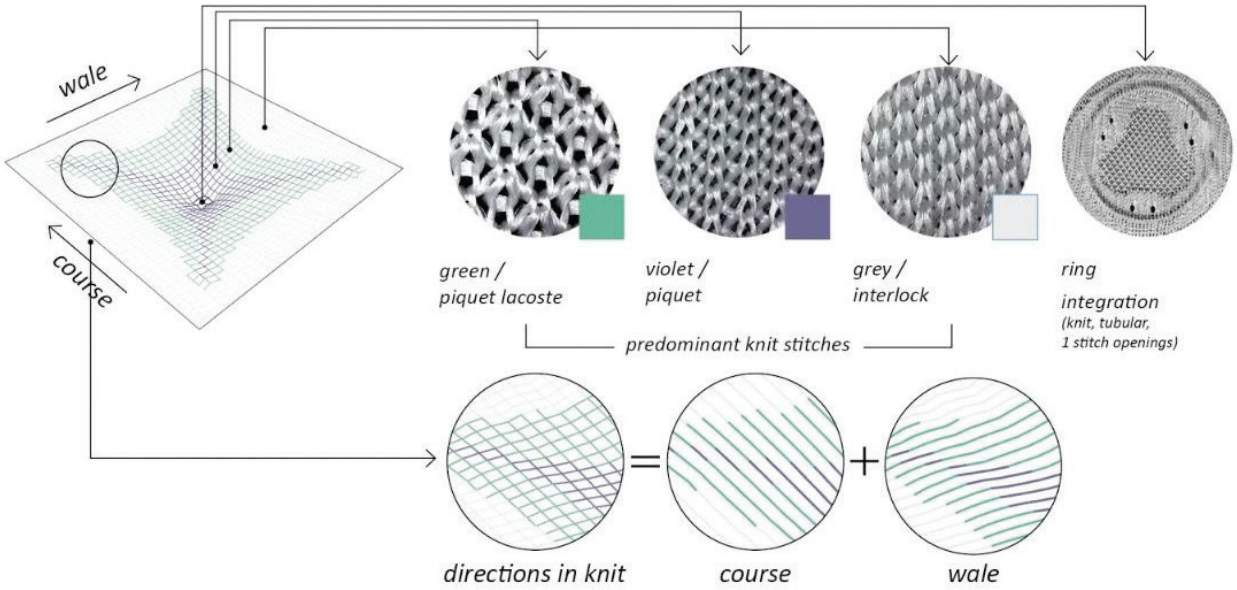




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2 Nine selected designs for the experiment.

3 Testing setup: graded CNC-knitted membrane tensioned to a suspended frame, loaded with 15 kg and scanned with FARO laser scanner.



4 Digital setup for graded knitted membranes: mesh bars are sorted by color and divided into groups by the direction of wales and courses. 4

METHOD FOR SIMULATION AND TESTING OF GRADED KNITTED TEXTILES

Digital and Physical Correlation

In this paper, we present the development of a simulation method for graded knitted membranes through the KnitVault experiment that is designed to validate the method by testing the correlation between the digital and the physical aspects of the exercise.

The digital tool aims to follow the differences of physical textile panel behavior by incorporating variegated knit properties through the digital mesh grading. Despite the same yarn being used for the entire membrane surface fabrication, intertwining it into the different stitches results in a diversified fabric stiffness. As a direct yarn-to-yarn stitch modeling is computationally heavy, the abstraction of the knitted surface into a quad-based mesh is a compromise (Fig. 4).

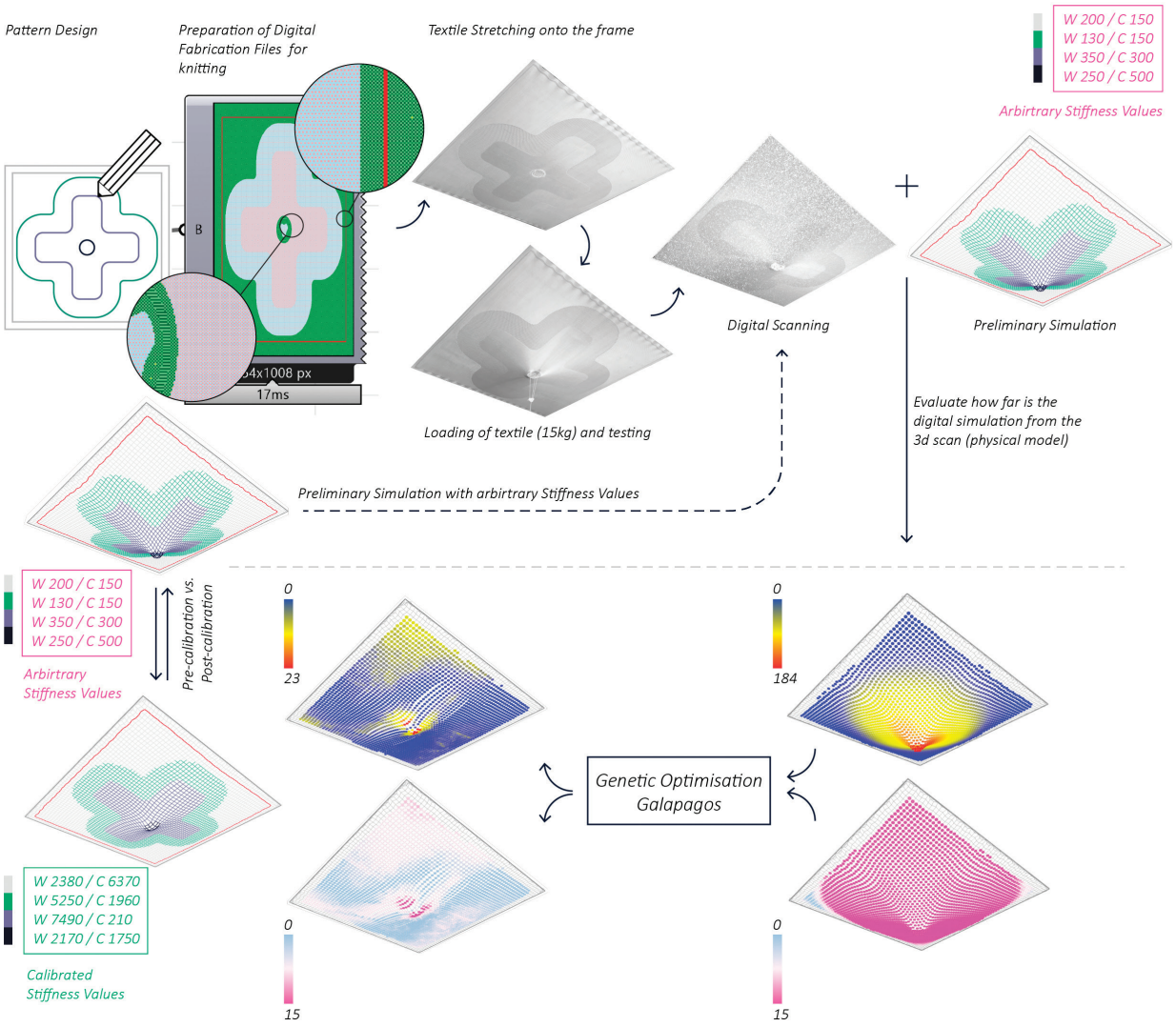
The physical side of the experiment is supported by the industrially knitted textile panels with varied topology patterns (Fig. 1), where three predominant stitch types (piquet, piquet lacoste, and interlock) are used to steer the surface degree of expandability under load (Fig. 4). Nine pattern configurations (Fig. 2), selected for the experiment, were designed manually to create different stitch-ratio proportions and pattern-drawing geometries. Although looking back to these decisions, the designs could be more irregular in order to challenge the experiment setup.

As the textiles are knitted on the industrial CNC-knitting machine using digitally generated bitmaps and equivalent machine settings throughout the production, the pattern difference is the only varied parameter between the panels. Once fabricated and attached to the frame, the dissimilarity in planar pretension is noticeable across the designs, which indicates the effect of patterns on the textile behavior.

Next, the panels are loaded with a 15 kg weight from the center, which results in different stretch depths, related to the pattern differences (Fig. 3). Each geometry is scanned with a FARO laser scanner in order to transfer the data into the digital environment. The processed point clouds of the loaded panels are compared to the corresponding simulations (K2 Engineering plug-in for Kangaroo [Brandt 2016]) for further digital tool calibration.

The directionality of the quad-mesh (UV arrangement) associates with the wale-course nature of knitted structures, where heterogeneous properties are defined through the mesh segments' stiffness assignment across the surface, defined by the pattern boundaries and the direction. The direction of knit is important to take into account, as it contributes to an anisotropic behavior of the fabric. Therefore, separate stiffness values are assigned to the course and wale directions of the mesh.

While the stiffness of the yarn is normally a known value, provided by the producer, *the stiffness of the knitted fabric made of this yarn remains unknown,* as various stitches can be used to form the textile. For that reason, it poses an interesting challenge to understand these relations in order to build a reliable, yet computationally viable, simulation tool of graded knitted fabrics.



5 Diagram of the workflow for graded knitted membrane design, digital setup, simulation, testing, and digital tool calibration. 5

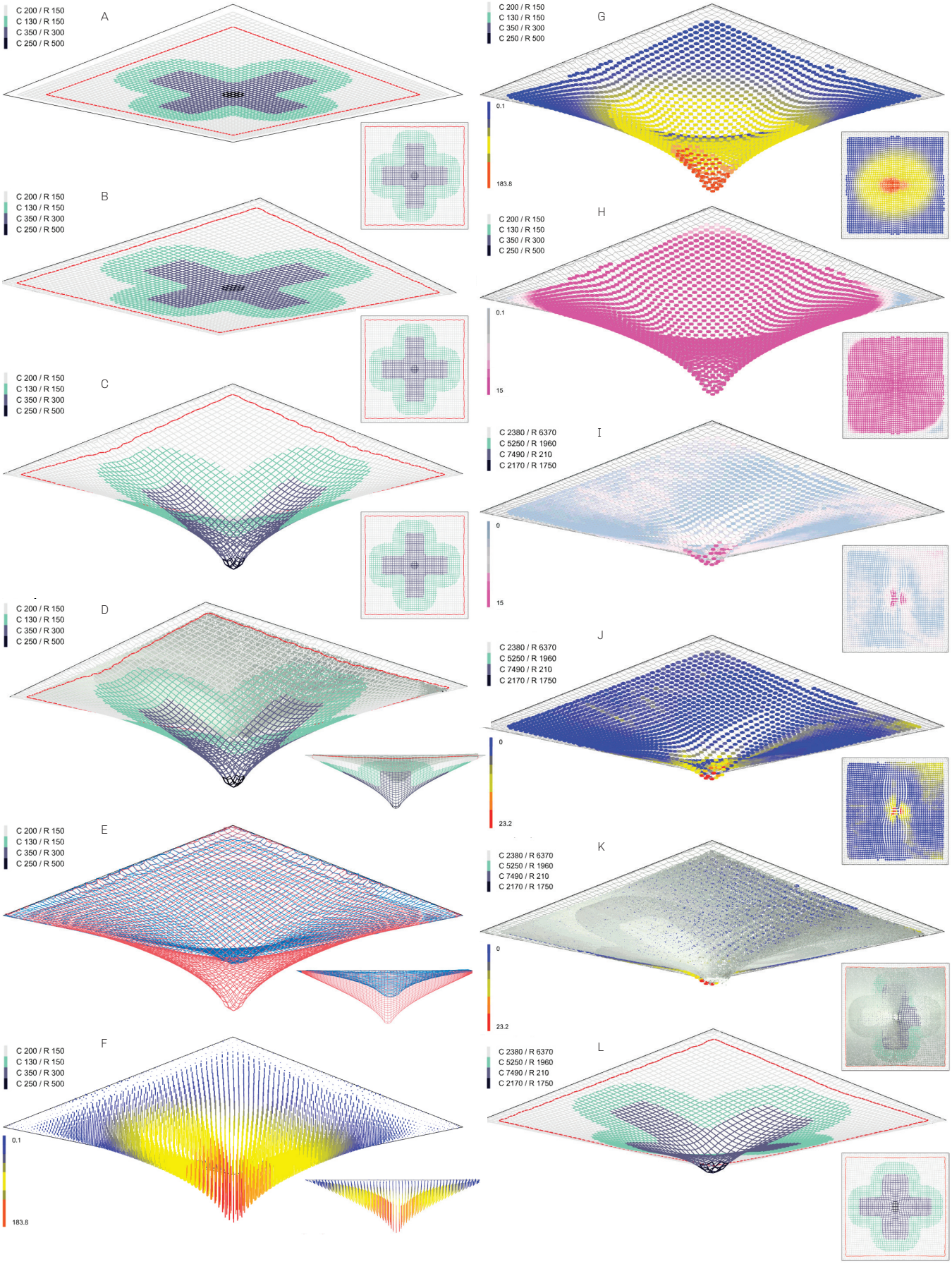
Digital Deviation Optimization

At first, when the real stiffness values are yet undiscovered, arbitrary numbers are assigned to the pattern zones in order to complete the simulation. This gives an arbitrary mesh geometry to work with.

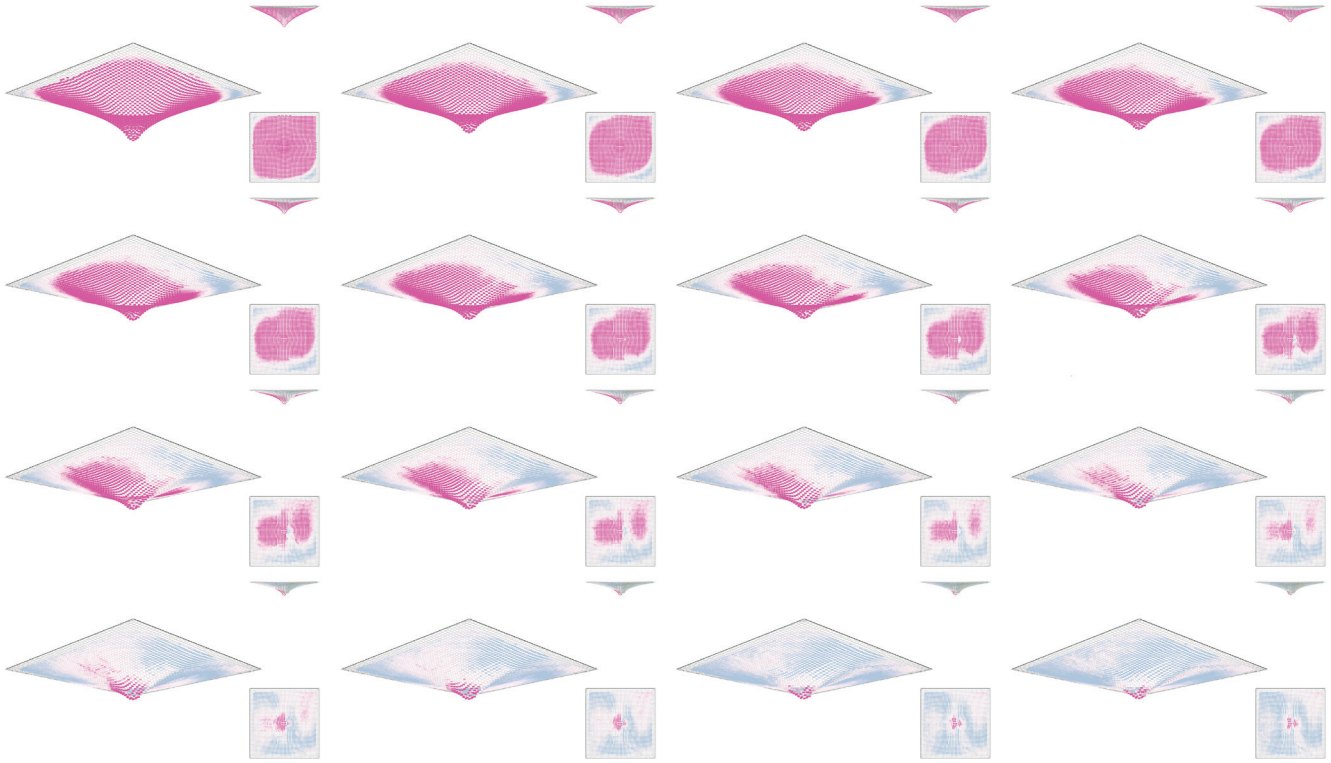
The simulation follows the same steps as in the physical installation of the fabric onto the test rig. In the beginning, the textile sample is stretched in plane with the testing frame and only then loaded with the weight. In the same way the pretension is introduced in the digital environment. These pretension values are then used in the second step of the simulation, when the pretensioned mesh is loaded with the 147 N in the z (down) direction (Fig. 6a, b, c). This allows us to build a more correct simulation of the three-dimensional geometry, corresponding more closely to the real behavior.

The point clouds are translated to the mesh geometry by projecting an arbitrary mesh (in red) to it and rebuilding the same topology mesh (in blue, Fig. 6e). This way there are two meshes with identical topologies that can be overlapped and compared. The comparison lies in calculating the distances between the mesh nodes (Fig. 6f), where it is visualized through the red-yellow-blue color mapping projected onto the simulation mesh (Fig. 6g). In the magenta mapping, the nodes are colored pink if the distance exceeds 15 mm (Fig. 6h).

At first, the mesh with the arbitrary values deviates a lot from the point cloud geometry (Fig. 6d). The prevalence of the magenta nodes (Fig. 6h) means a high degree of discrepancy between the simulation and the equivalent physical geometry. The aim of the task is to lower the distance between the red and the blue mesh by achieving



6 Step-by-step actions of graded membrane stiffness value optimization membrane design, digital setup, simulation, testing, and digital tool calibration



7 Incremental improvement of the stiffness value set by reduction of deviation value between the simulated mesh and the point cloud. 7

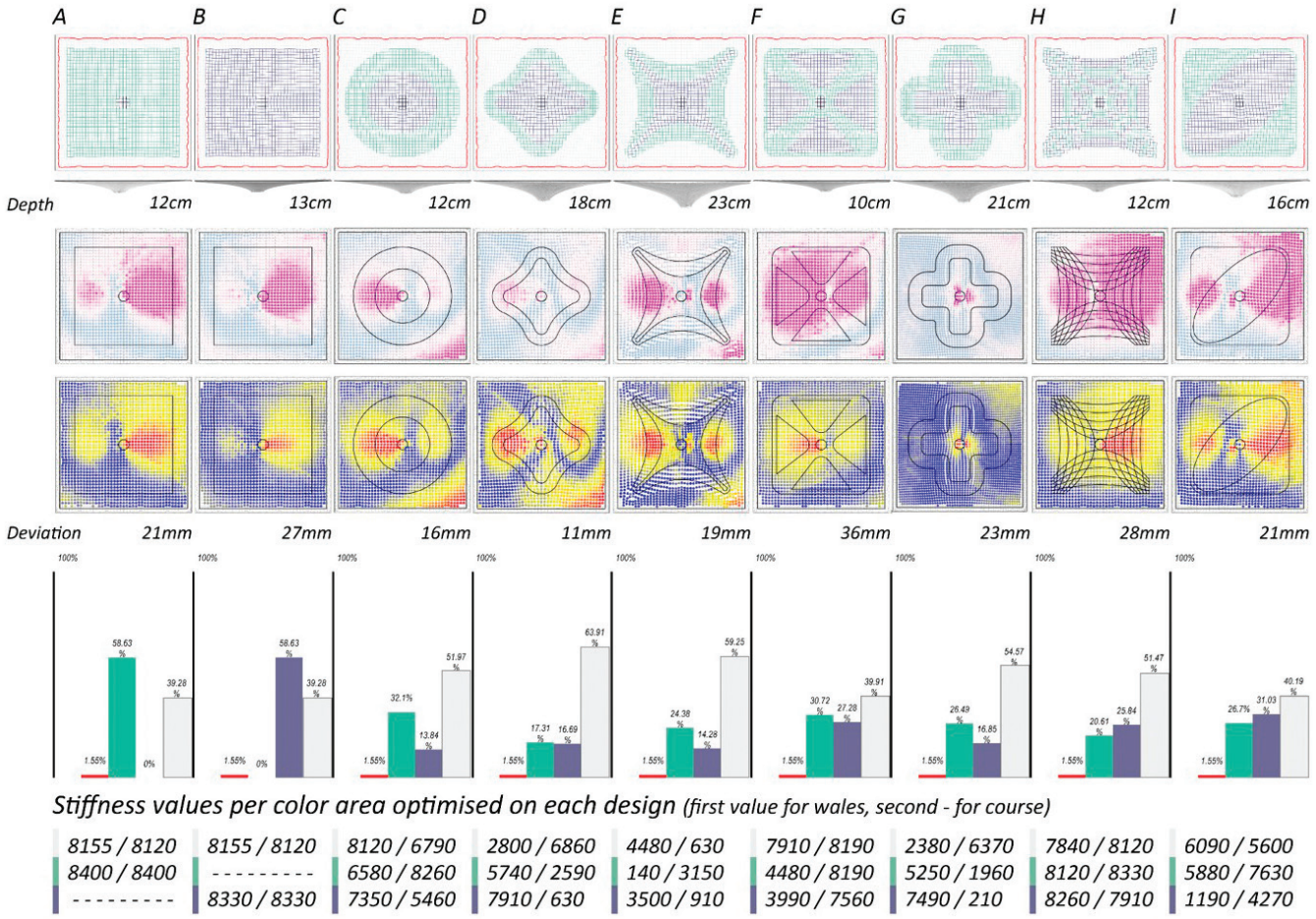
the smallest maximum distance and reducing the number of magenta nodes in the simulated mesh.

To derive the set of unknown stiffness values, a genetic optimization algorithm is used to minimize the distance between the digital mesh and the point cloud of the corresponding textiles (Fig. 5). The optimization loop is established with the Galapagos plug-in for Grasshopper Rutten 2020), integrated in a double solver simulation with Kangaroo (Piker 2017) and K2 Engineering workflows. Here the engine is going over the stiffness value combinatorics within the given boundaries (150–8500 MPa) and applying them to the mesh zones. With each stiffness value application, the algorithm is measuring the distance from the resulting simulated mesh to the point cloud. If it exceeds the maximum desired value, the search for the optimal stiffness values continues until the deviation is acceptably low. On average, the search for each design solution took 10–15 hours and an average of 20,000–25,000 iterations. Figure 7 shows the incremental reduction of both magenta nodes and the maximum distance, while the mesh is getting closer to the point cloud geometry.

RESULTS AND DISCUSSION

This paper describes the method of stiffness value determination for graded knitted membrane material. Initially, unknown mesh stiffness values are optimized, using corresponding point clouds as target geometries for minimizing the distance to the mesh. As a result of the exercise, the stiffness values that would bring the digital mesh very close to the corresponding physical model geometry have been found (Fig. 8). The tolerance between the point cloud and the simulated mesh is between 16 mm and 36 mm, which is acceptable for the expandable fabrics. Although the results for each geometry were very promising, the stiffness values for the same fabric type differ from design to design, and the cross-application of discovered values would give deviating results. This means, for example, that the stiffness values that work perfectly for design D would not work for F. This challenge provoked the authors to rethink the method and to find a potential system in the results to improve the method for the future development of the work.

For that reason all the discovered stiffness value sets are cross-applied to all the other designs. This exercise is explained in Figure 9 through D-, C-, and E-based optimized results. The results for each cross-application are sorted based on the deviation from the corresponding point cloud: from the smallest in the target design to the most deviating in the rest of them. When looking at the selection of design pairs based on their stitch ratio proportion similarity (D/G, C/E, I/H, E/H), interesting findings emerged. In the analysis of the results, the following aspects are taken into account:



8 Diagram showing (top to bottom) the tested design range, optimization results mapped onto the mesh, stitch ratio proportions, and stiffness values.

stitch ratio, geometry of the pattern, and the depth of the cones under the stretch.

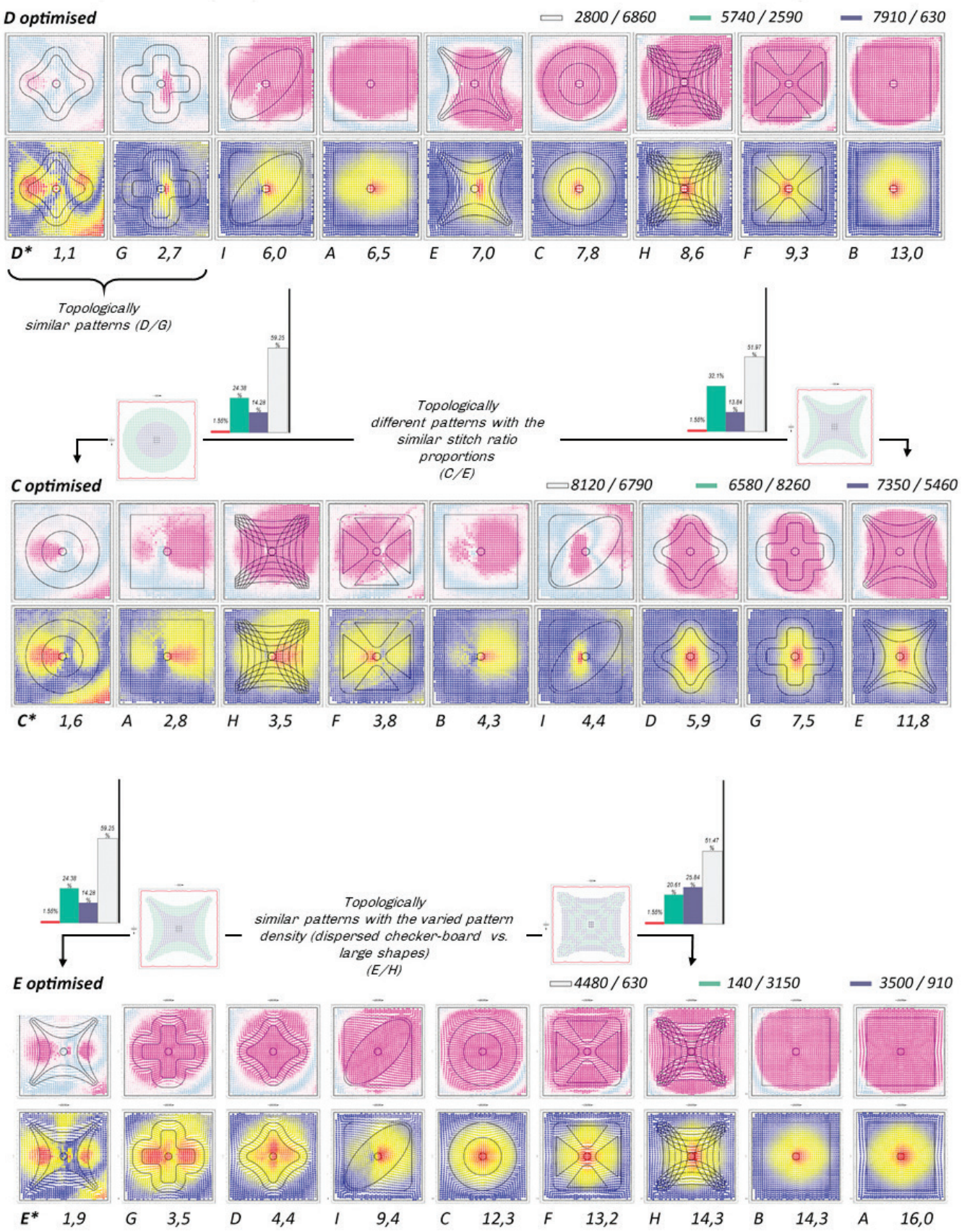
Cross-application of stiffness values between topologically similar patterns ("cross-like" in D/G) shows less deviation from the equivalent point clouds (Fig. 9 top), while topologically different patterns, although with similar stitch ratio (C/E), resulted in a larger error to the point clouds (Fig. 9 middle). Topologically similar designs with a close stitch ratio (E/H) but variation in pattern density (dispersed checkerboard vs. large shapes) performed a noncompatibility of stiffness values. The numbers resulting from optimization on the shallower designs (A, B, C, F, H) are higher, while the values trained on deeper geometries (D, E, G) are lower (Fig. 10). This is logical, as the stiffer mesh has more resilience against the load and therefore results in a shallower cone.

Overall, the value of discovered stiffness values remains doubtful, as it is unreliable to reuse them for pattern geometry other than the one that was used to train the optimization. The discrepancies among the stiffness values can be explained by the lack of pattern boundary tracing. As the pattern linework was not traced in the point clouds, it is challenging to evaluate whether the simulated mesh has taken the exact right shape. Possibly, while the overall mesh boundaries were quite close after optimization, the inner boundaries of the pattern might remain off.

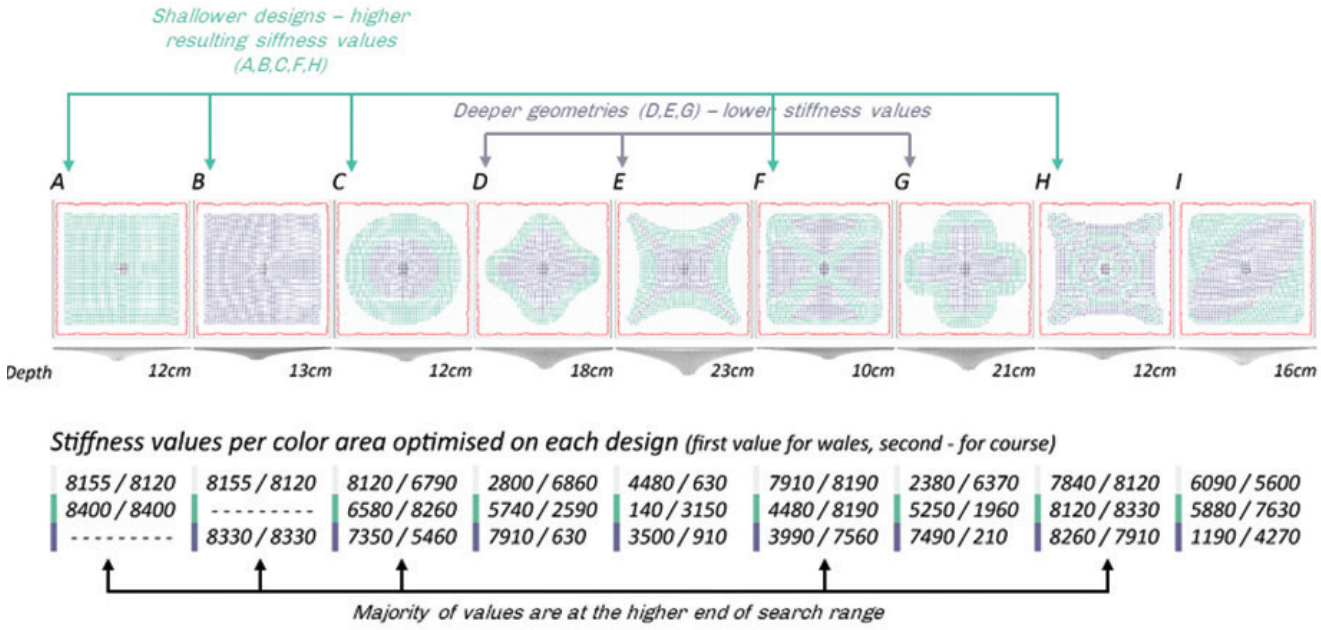
Additionally, in the described optimization setup, the relations between the stitch type areas were not defined but rather left too open. The same stiffness value range condition (150–8500 MPa), introduced to all the zones, created a situation where the same stitch type zone is obtaining both the low and the high stiffness values in the different designs.

REFLECTION AND CONCLUSION

Despite the challenges encountered in the universality of the method, the presented workflow is an important step toward an understanding of the graded knit behavior. The optimization of stiffness values per design showed a high degree of precision, interlinking the pattern design, fabrication data, physical testing data, and the corresponding digital simulation.



9 Results of the cross-application of the stiffness values trained on a particular design.



10 The resulting stiffness value sets for each panel design. 10

A future research work would imply a further increase of digital complexity, introducing a wider search range of stiffness values, finding a way to quantify the pattern density (dispersed vs. large zones), establishing the relations between the stitch types (stiff vs. expandable), and introducing additional springs for reflecting the stitch type differences. These implementations might potentially increase the method's precision, transforming it to the tool for generating stiffness values for heterogeneous knitted membranes.

A more reliable digital tool for graded knitted membranes would allow for a greater confidence in design with this challenging but high potential material for architectural applications. Predictability of graded knit material behavior would build trust in using it for large-scale architectural elements, where heterogeneous material density and elasticity can contribute to functionality and structural performance while being highly sustainable and waste-free in manufacturing.

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IMAGE CREDITS

All images and drawings are by the authors.

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Mette Ramsgaard Thomsen examines the intersections between architecture and new computational design processes. During the last 15 years her focus has been on the profound changes that digital technologies instigate in the way architecture is thought, designed, and built. In 2005 she founded the Centre for IT and Architecture research group (CITA) at the Royal Academy of Fine Arts, School of Architecture, Design and Conservation, where she has piloted a special research focus on the new digital-material relations that digital technologies bring forth. Investigating advanced computer modeling, digital fabrication, and material specification, CITA has been central in forming an international research field examining the changes to material practice in architecture.